If we're stuck on one world, we're limited to a single case; we don't know what else is possible. Then—like an art fancier familiar only with Fayoum tomb paintings, a dentist who knows only molars, a philosopher trained merely in Neo-Platonism, a linguist who has studied only Chinese, or a physicist whose knowledge of gravity is restricted to falling bodies on Earth—our perspective is foreshortened, our insights narrow, our predictive abilities circumscribed. By contrast, when we explore other worlds, what once seemed the only way a planet could be turns out to be somewhere in the middle range of a vast spectrum of possibilities. When we look at those other worlds, we begin to understand what happens when we have too much of one thing or too little of another. We learn how a planet can go wrong. We gain a new understanding, foreseen by the spaceflight pioneer Robert Goddard, called comparative planetology.

The exploration of other worlds has opened our eyes in the study of volcanos, earthquakes, and weather. It may one day have profound implications for biology, because all life on Earth is built on a common biochemical master plan. The discovery of a single extraterrestrial organism—even something as humble as a bacterium—would revolutionize our understanding of living things. But the connection between exploring other worlds and protecting this one is most evident in the study of Earth's climate and the burgeoning threat to that climate that our technology now poses. Other worlds provide vital insights about what dumb things not to do on Earth.

Three potential environmental catastrophes—all operating on a global scale—have recently been uncovered: ozone layer depletion, greenhouse warming, and nuclear winter. All three discoveries, it turns out, have strong ties to the exploration of the planets.

(1) It was disturbing to find that an inert material with all sorts of practical applications—it serves as the working fluid in refrigerators and air conditioners, as aerosol propellant for deodorants and other products, as lightweight foamy packaging for fast foods, and as a cleaning agent in microelectronics, to name only a few—can pose a danger to life on Earth. Who would have figured?